

SHUVAM PAL

+91 85830 54679 ✉ ishuvam.pal@gmail.com  Shuvam Pal  unichronic

Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions

Remote

- Built a resilient **Puppeteer/Node.js automation engine** for MCA filings, reducing manual filing time by **85%** through retries, network-idle monitoring, and recoverable browser-state handling.
- Optimized an **AI-assisted company-registration pipeline** for unique company-name suggestions, descriptions, and filing-field drafts, reducing manual drafting/rework across MCA workflow preparation.
- Containerized services with **Docker**, deployed to **Google Cloud Run**, and automated releases with **GitHub Actions**, improving deployment velocity by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalio: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built MRI segmentation tooling for **95 anatomical regions** and integrated model outputs into visualization workflows that needed repeatable, inspectable processing.
- Profiled volume-processing paths and reduced runtime by **87%** with async/parallel execution and fewer repeated post-processing passes.
- Improved 3D reconstructed surface visualization flows, keeping image-processing outputs usable for debugging and review.

Open Source

[Contributions](#) 

- Contributed to Invesalio, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, adapting fixes to maintainer review and existing observability/debugging practices.

Projects

Postificus | *Go, Echo, Rod, PostgreSQL, Redis, RabbitMQ, Docker, AWS, Prometheus*

[GitHub](#) 

- Built observable Go services for a **content distribution platform** using browser automation, with RabbitMQ workers, DLQs, retries, Redis autosave, and PostgreSQL persistence.
- Added circuit-breaker logic and failure isolation for third-party publishing paths, keeping browser automation failures observable and recoverable instead of cascading through the system.

Murdoc | *Python, FastAPI, MCP, OpenTelemetry, Prometheus, Docker*

[GitHub](#) 

- Instrumented an AI gateway with **OpenTelemetry metrics/traces**, Prometheus health checks, route profiles, and audit logs tying model/tool requests to policy outcomes.
- Built replayable decision records for prompt, tool, retrieval, and output layers so high-risk agent behavior could be correlated with request context and failure modes.

Swish Support System | *Python, Go, FastAPI, Gin, PostgreSQL, Redis, Langfuse*

[GitHub](#) 

- Built AI-assisted backend workflows for order status, delays, refunds, trust checks, and escalation using tool calls over operational APIs with PostgreSQL and Redis state.
- Added retrieval-backed responses, confidence thresholds, human handoff paths, conversation memory, and traceable action records for ambiguous or high-risk user workflows.

Technical Skills

Languages: Go, Python, TypeScript

Frameworks: FastAPI, Echo, Gin, Node.js

Tools: OpenTelemetry, Prometheus, Langfuse, PostgreSQL, Redis, RabbitMQ, Docker, GitHub Actions